

User Trend Analysis to Increase Membership Conversion

Identified Through BigQuery and Visualized in Data Studio

Executive Summary

The analysis compares differences between how casual riders and annual members utilize Cyclistic bicycles to design a marketing strategy that focuses on converting casual riders into annual members. Key differences in time and location in user trends indicate that promotional campaigns should occur during the weekends near the Navy Pier and DuSable Lake Shore Dr./Monroe St. stations when casual riders are most likely to be active, and special summer promotions should occur from June to August when ridership peaks.

Business Problem

Annual members are more profitable than casual riders, so Cyclistic needs to pivot their marketing strategy from relying on building a general awareness and appealing to broad consumer segments to a focused strategy that incentivizes casual riders to convert to annual membership.

The following analysis serves as a basis for the Marketing Team's new approach by revealing when and where promotional events should occur to reach the most casual riders as they are actively using Cyclistic bicycles.

Data Source Description

The data is collected from Cyclistic user data within the past 12 months (July 2025 to June 2026) from the internal database. The data was downloaded as 12 CSV files (one file per month) then promptly uploaded into BigQuery for cleaning and analysis; the total number of rows of data is approximately 6 million.

It has 3 main limitations: it does not explicitly include trip information for assistive bike models, approximately 20% of trip records do not document start and/or end station for electric bikes, and approximately 2% trip lengths are invalid. These limitations are addressed, minimizing the impact on the analysis completed when possible.

For a complete definition of all fields, transformations, and null handling decisions, refer to Appendix A: Data Dictionary.

Methodology

Given the volume of data, the CSV files were combined into a dataset in BigQuery. The data was reviewed and cleaned (Queries 1.1 to 1.6) by first converting ride length format to integers that represent total seconds of trip and extracting the Day of Week the trip was conducted into a separate column for data analysis.

Query 2.1 returned the average and maximum ride length in seconds between casual rides and annual members and the most popular day of the week for each member type to take a trip with Cyclistic bikes.

After reviewing the calculations, null values in the start and/or end station IDs and names were identified. Query 1.7 checks the quantity of null values for the start and/or end station IDs and names to verify how significant these null values are. Query 1.8 checks if there is a relationship between missing station ID and names and bike type based on Start Station ID. (Note: it was not necessary to check end station values since the amount of start station null values for electric bikes matched the total amount of start station null values.)

These null values consisted approximately 20% of the data, so these trip records remained in the analysis but is a limitation of the findings for users.

Following the discovery of these null values, Queries 2.2 to 2.4 checked for invalid ride lengths (trip times that were under 1 minute or over 24 hours); these outliers in ride length only consisted of approximately 2% of the data and consequently excluded from further analysis by creating a new table in Query 3.1.

The remaining queries calculate average ride length, most active day of the week and months, and most popular stations per month for casual riders and annual members to identify differences in how each rider type utilizes Cyclistic bikes.

Skills and Tools Used

BigQuery was selected as the platform to perform SQL queries to clean and analyze the data due to tool availability and prior experience. A variety of queries ranging from UNION ALL to combine fields, EXTRACT to pull specific values, calculations like AVG and MAX, and CTEs for more complex tasks were used to thoroughly examine user trends by rider type.

Data Studio (formerly Looker Studio) was selected as the visualization tool because of the seamless integration with BigQuery. Pie charts, bar graphs, and time series graphs were identified as the best visualizations to display the makeup of Cyclistic's user population and changes within the past 12 months.

For a complete summary of SQL queries, refer to Appendix B: SQL Queries.

Figure 1: Total Trip Counts Per Day of Week by Rider Type

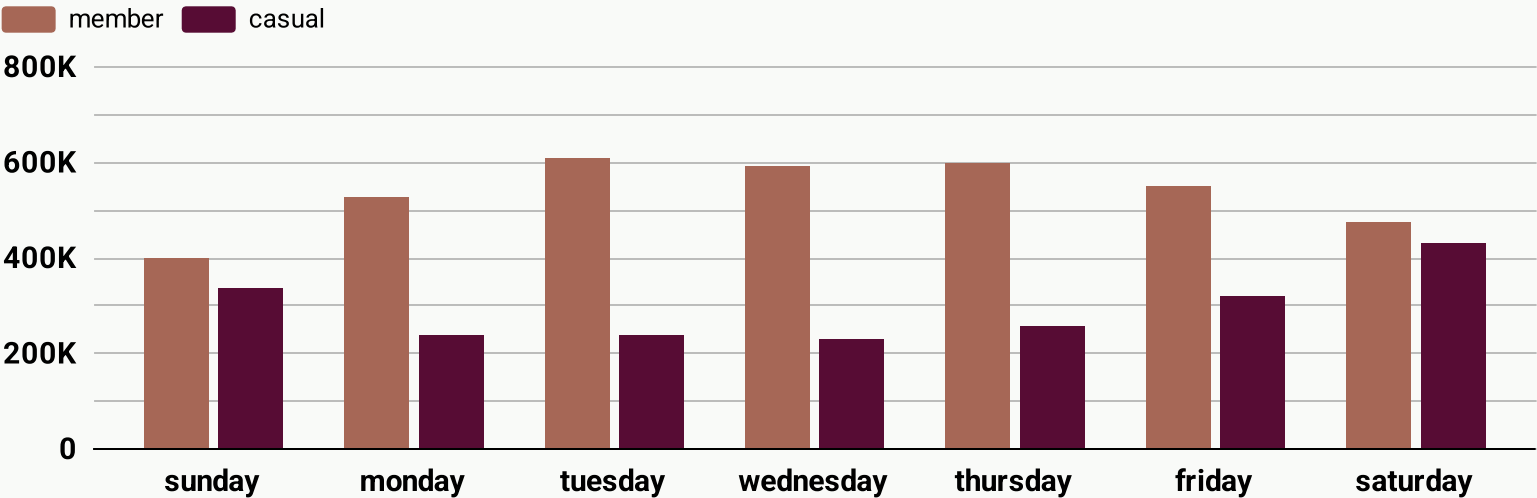


Figure 1 displays which days of the week each rider type is most active.

Analysis, Findings, and Recommendations

The findings identified and modeled indicate that promotional campaigns should occur during the weekends near the Navy Pier and DuSable Lake Shore Dr./Monroe St. stations when casual riders are most likely to be active, and special summer promotions should occur from June to August when ridership peaks.

Recommended Marketing Strategy 1: Promotional campaigns should occur during the weekends when Casual Riders are most likely to be active.

According to Figure 1: Total Trip Counts Per Day of Week by Rider Type, Annual Members mostly use bikes throughout the work week while Casual Riders usage peaks during the weekend.

Marketing initiatives to convert users from Casual Riders to Annual Members should occur when they are most likely to be active, so their exposure to Cyclistic advertising is increased.

Recommended Marketing Strategy 2: Promotional campaigns should occur mainly at Navy Pier and DuSable Lake Shore Dr./Monroe St. where Casual Riders are most likely to be active.

Approximately 74% of bikes used by Casual Riders were docked at either Navy Pier or DuSable Lake Shore Dr./Monroe St at the start and/or end of their trip as shown in Figure 2.1: Most Popular Stations Frequented by Casual Riders.

There is no overlap between the most popular stations for Casual Riders and Annual Members (refer to Figure 2.2: Most Popular Stations Frequented by Annual members). Therefore, marketing initiatives should be conducted primarily at these two locations to target the highest percentage of Casual Riders while they are active.

Concentrating marketing initiatives at these times and locations shifts Cyclistic's marketing strategy from relying on building a general awareness and appealing to broad consumer segments to a focused strategy that incentivizes casual riders to convert to annual membership.

Figure 2.1: Most Popular Stations Frequented by Casual Riders

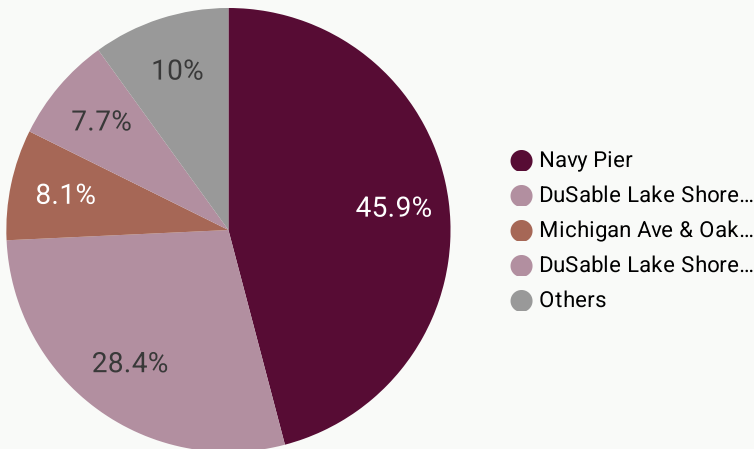


Figure 2.1 shows the percentages of the most popular stations frequented by Casual Riders.*

Figure 2.2: Most Popular Stations Frequented by Annual Members

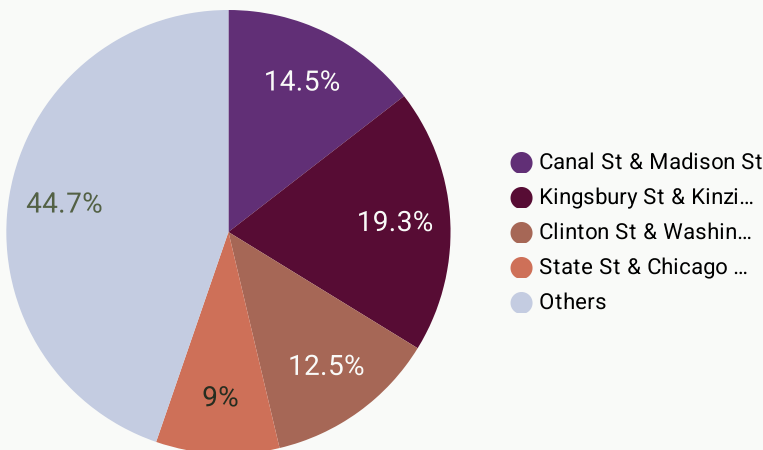


Figure 2.2 shows the percentages of the most popular stations frequented by Annual Members.*

*This data is only representative of the 80% of records that did not contain null values for the start/end station ID or name.

Figure 3: Total Trips Per Month By Rider Type

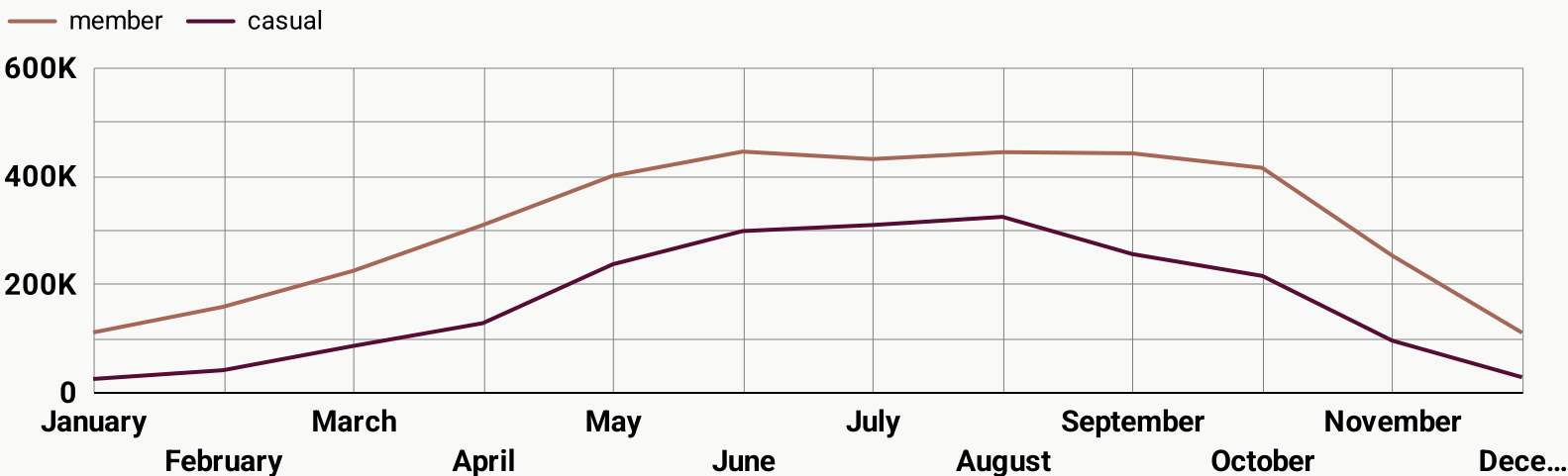


Figure 3 displays the total trips Cyclistic users took per month by rider type from July 2025 to June 2026.

Recommended Marketing Strategy 3: Promotional campaigns should be targeted during the summer months of June to August when ridership peaks.

Both Casual Riders and Annual Members ridership peaks to minimum 440k monthly Annual Member users and approximately 300k monthly Casual Riders in the summer time, specifically from June to August (Figure 3: Total Trips Per Month By Rider Type).

Cyclistic must time marketing initiatives during this period of the year when there are more users, promoting annual membership benefits and/or offering discount rates for the summer time from June to August.

Limitations

As discussed, the data has 3 main limitations: it does not explicitly include trip information for assistive bike models, approximately 20% of trip records do not document start and/or end station for electric bikes, and approximately 2% trip lengths are invalid.

There is no explicit field that identifies if a rideable type is an assistive model, therefore the analysis assumes all trips recorded are of users who did not utilize an assistive model. The lack of verifiable data for assistive model users limits the outreach Cyclistic can do because there is no information on the popularity of these rideable options.

Furthermore, approximately 20% of records do not document start and/or end station for electric bikes. This is a significant portion of the data and could not be excluded from the analysis without greatly skewing the possible findings.

The approximately 2% of records with invalid trip lengths of less than 1 minute or over 24 hours is also an issue since it affects the accuracy of reflecting how long users use Cyclistic bikes. Although, this was a small portion of the data and could be excluded without greatly affecting the analysis.

Next Steps

First, the previously discussed limitations of the analysis must be addressed by identifying how trip data is incorrectly recorded in Cyclistic's system and including a field that identifies if a user is using an assistive rideable type. Correcting these errors now improves the accuracy of future data collection and analysis.

The suggested changes in marketing strategies should be tracked for a full calendar year, ensuring an accurate assessment of targeted promotional campaigns, especially in the summer time. Additional research should determine why users would want or need an annual membership to further hone marketing campaigns that appeal to the specific reasons Casual Riders may convert to Annual Members. This can be conducted by surveying casual riders at Navy Pier and DuSable Lake Shore Dr./Monroe St. during peak summer weekends.

Furthermore, Annual Members comprise 64.8% of Cyclistic's user population, and they ride at least 37% more than Casual Riders throughout the year (Figure 3: Total Trips Per Month by Rider Type). So, specific investigations on Annual Member users and their preferences should be conducted to ensure continued satisfaction. For example, both rider types prefer electric over classic bikes with electrics used 66.8% of the time, so ensuring electric bike availability for members at most popular stations during peak times to maintain retention of users who are already annual members.

Appendix A: Data Dictionary

Cyclistic User Trend Analysis to Increase Membership Conversion | Google Data Analytics Capstone Project

Source: Cyclistic User Data, July 2025 – June 2026 | Raw records: 5,932,349 | Clean analytical records: 5,763,158

Source Fields

Original columns from the raw Cyclistic dataset.

Field Name	Data Type	Description	Possible Values	Transformation	Null Handling
ride_id	STRING	Unique identifier assigned to each individual ride	Alphanumeric string	None	No nulls expected
rideable_type	STRING	Type of bike used for the ride	classic_bike, electric_bike	None	No nulls expected
started_at	TIMESTAMP	Date and time the ride began	Jul 2025 – Jun 2026	None	No nulls expected
ended_at	TIMESTAMP	Date and time the ride ended	Jul 2025 – Jun 2026	None	No nulls expected
start_station_name	STRING	Name of the station where the ride began	Station name or NULL	None	~21% null, electric_bike only. Retained; excluded from station analysis.
start_station_id	STRING	Unique ID of the departure station	Alphanumeric ID or NULL	None	~21% null, mirrors start_station_name nulls exactly
end_station_name	STRING	Name of the station where the ride ended	Station name or NULL	None	~22% null, electric_bike only. Retained; excluded from station analysis.
end_station_id	STRING	Unique ID of the arrival station	Alphanumeric ID or NULL	None	~22% null, mirrors end_station_name nulls exactly
start_lat	FLOAT	Latitude coordinate of the ride start location	Decimal degrees	None	May be null where station data is absent
start_lng	FLOAT	Longitude coordinate of the ride start location	Decimal degrees	None	May be null where station data is absent
end_lat	FLOAT	Latitude coordinate of the ride end location	Decimal degrees	None	May be null where station data is absent
end_lng	FLOAT	Longitude coordinate of the ride end location	Decimal degrees	None	May be null where station data is absent
member_casual	STRING	Indicates whether the rider holds an annual membership or is a casual pay-per-ride user	member, casual	None	No nulls expected

Derived Fields

Calculated columns added during data preparation in BigQuery.

Field Name	Data Type	Description	Possible Values	Transformation	Null Handling
ride_length	INTEGER	Duration of the ride in seconds	60 to 86,400 after cleaning	TIMESTAMP_DIFF(ended_at, started_at, SECOND)	Rides ≤60s, ≥86,400s, and negative excluded. 167,814 rows removed total.
day_of_week	INTEGER	Day of the week the ride started as an integer	1=Sunday through 7=Saturday	EXTRACT(DAYOFWEEK FROM started_at)	No nulls expected
month	INTEGER	Month the ride started as an integer	1=January through 12=December	EXTRACT(MONTH FROM started_at)	No nulls expected
day_name	STRING	Day of week as a readable string derived from day_of_week	sunday through saturday	CASE day_of_week WHEN 1 THEN 'sunday' ... END	No nulls expected

NOTES

- Station null values (~21% start, ~22% end) are exclusive to electric_bike rides. Root cause was not confirmed in available documentation. Rows were retained to avoid bias but excluded from station-level analysis.
- No data is available identifying riders who use adaptive or assistive equipment, limiting demographic inclusivity analysis.
- Ride length outliers excluded: 29 negative rides, 162,246 rides under 60 seconds, and 5,539 rides over 86,400 seconds. Total excluded: 167,814 rows.
- Data covers July 2025 through June 2026. Seasonal patterns may not generalize beyond this 12-month window.

Appendix B: SQL Query Summary

Cyclistic User Trend Analysis to Increase Membership Conversion | Google Data Analytics Capstone Project

Source: Cyclistic User Data, July 2025 – June 2026 | Tool: BigQuery (Google Cloud SQL) | Total queries: 16

Table Management

Function / Clause	Category	Purpose	Application in Project
CREATE TABLE AS	Table Management	Saved every analytical output as a permanent table in the data_processed dataset for use in Data Studio	Used across all 16 queries to persist results
CREATE OR REPLACE TABLE	Table Management	Overwrote an existing table without manually dropping it first	Used when correcting outlier exclusion logic
DROP TABLE	Table Management	Permanently deleted a table to allow recreation with corrected logic	Used to reset the data_no_outliers table

Data Combination

Function / Clause	Category	Purpose	Application in Project
UNION ALL	Set Operation	Stacked all 12 monthly trip tables vertically into one unified dataset, preserving all rows including duplicates	Combined 5,932,349 rows from 12 separate source tables

Data Transformation

Function / Clause	Category	Purpose	Application in Project
TIMESTAMP_DIFF()	Timestamp Function	Calculated the duration of each ride in seconds by subtracting started_at from ended_at	Generated the ride_length column
EXTRACT()	Date Function	Extracted the day of week and month from the started_at timestamp as integers	Generated day_of_week (1–7) and month (1–12) columns
FORMAT()	String Function	Converted integer seconds into HH:MM:SS readable format for display	Applied to AVG(ride_length) in mean ride length summary
CAST()	Type Conversion	Converted FLOAT ride length values to INT64 for use inside FORMAT() and MOD() functions	Required for HH:MM:SS conversion formula
CASE WHEN	Conditional Logic	Converted integer day_of_week values into readable day name strings	Generated day_name column (sunday through saturday)

Aggregate Functions

Function / Clause	Category	Purpose	Application in Project
COUNT(*)	Aggregate	Counted total rides per group — by rider type, day, month, station, and bike type	Core metric across all segmented analysis tables

Function / Clause	Category	Purpose	Application in Project
COUNTIF()	Conditional Aggregate	Counted rides matching a specific condition, used to pivot member and casual counts into side-by-side columns	Used in percentage difference queries
AVG()	Aggregate	Calculated mean ride length across all rides and by rider type	Used in descriptive stats and mean ride length tables
MAX()	Aggregate	Identified the longest single ride in the dataset overall and by rider type	Used in descriptive stats and max ride length tables
ROUND()	Numeric Function	Rounded percentage difference results to 2 decimal places for readability	Applied to percentage difference calculations
MOD()	Numeric Function	Extracted remainder values for minutes and seconds in HH:MM:SS conversion	Used inside FORMAT() for ride length display

Filtering and Cleaning

Function / Clause	Category	Purpose	Application in Project
WHERE	Filter Clause	Excluded outlier rides (negative, under 60 seconds, over 86,400 seconds) and null station values from relevant analyses	Applied in data_no_outliers and station queries
IS NOT NULL	Null Filter	Excluded rows with missing station names from station frequency analysis	Applied in both start and end station CTEs
COUNTIF(IS NULL)	Null Audit	Quantified the scale of null station values across start and end station columns to inform handling decision	Revealed ~21% null rate exclusive to electric bikes

Common Table Expressions (CTEs)

Function / Clause	Category	Purpose	Application in Project
WITH ... AS ()	CTE	Created temporary named result sets within a single query to break complex logic into readable sequential steps	Used in station frequency, mode of day, and top 10 station queries
Multiple CTEs (chained)	Chained CTE	Chained up to four CTEs in sequence — start station, end station, combined, ranked — each building on the previous result	Used in top 10 stations per month query

Window Functions

Function / Clause	Category	Purpose	Application in Project
ROW_NUMBER() OVER (PARTITION BY)	Window Function	Ranked stations by ride count within each member_casual and month group independently, enabling top N filtering per segment	Used to identify top 10 stations per rider type per month
SUM() OVER ()	Window Function	Calculated the total ride count across all groups without collapsing rows, enabling percentage share calculation per group	Used in ride percentage share query

Subqueries

Function / Clause	Category	Purpose	Application in Project
Scalar Subquery	Subquery	Embedded a self-contained SELECT inside the outer SELECT to calculate mode of day_of_week independently from global AVG and MAX calculations	Used in descriptive statistics query

Grouping and Sorting

Function / Clause	Category	Purpose	Application in Project
GROUP BY	Grouping	Aggregated rows by one or more categorical columns to produce summary statistics	Applied in all 16 analytical queries
ORDER BY ... DESC	Sorting	Sorted results by ride count in descending order to surface highest-frequency values at the top	Applied in station and day of week queries

NOTES

- All queries were written and executed in Google BigQuery using standard SQL dialect.
- Data was sourced from Cyclistic User Data (12 monthly files) and combined using UNION ALL.
- Final analytical tables were connected to Data Studio (formerly Looker Studio) for dashboard visualization.
- All SQL queries are saved in the project GitHub repository under /queries.

Disclaimers

This report was completed in August 2026 as the capstone project for the Google Data Analytics Professional Certificate offered through Coursera. The dataset was provided as part of the course curriculum and is based on publicly available data. All analysis, SQL queries, visualizations, and written content were independently produced by Margo Maliwat. This report is intended for portfolio and professional development purposes only. The findings and recommendations are based solely on the data provided and do not represent the views or strategies of any organization.